

t-SNE Problem Boot Camp

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December 5, 2024

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1 Summary and Overview

This section is used to summarize the main idea and process of the paper. This is a good process for learning and it would be beneficial to check with Tuca whether the understanding of the big picture is correct. Later in this section we will highlight the main difference between this document and the original paper should the readers are looking for the lemmas associated with the general output kernel.

From the first look the original paper [1] is daunting and painful to read because there are many definitions, one built upon another. Therefore tracking the computation in such settings can be exhausting. However, after going through the entire thing one can find the underlying logic of the paper is very simple and natural. The ultimate goal is to prove the convergence of $L_{n,\rho}(X,Y)$. More specifically, how can we convert this quantity to its continuous version, with the empirical measure replaced by the limiting measure and the summand replaced by a continuous integrand. We cannot just do it at once, but one at a time, and lower bound/upper bound separately.

Difference from the Original Paper

Most of work starts after section 5. Before that, some proofs are provided either because the original paper omitted them or the original paper had minor mistakes. The key difference between the original kernel and this notes is to write down

$$\frac{\partial L_{n,\rho}(X,Y)}{\partial y_i} = - \sum_{j=1, j \neq i}^n (p_{ij} - q_{ij}) \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)}. \quad (1.1)$$

This actually gives us a clearer idea why the set $\tilde{\mathcal{P}}$ takes its form in the original paper. Also we require the general kernel to have bounded derivative because that is essential for equicontinuity in lemma 18. After realizing the key differences the rest of the proofs will come natural.

2 Progress and Roadmap for Ph.D Thesis

This section is used to lay out the plan for next steps and how we could finish the Ph.D thesis on time. At this point, I have a very good understanding of the paper and from the computations in this notes, we know that the generalization of the output kernel does NOT interfere with the input kernel. Therefore, this notes serve more than just a warmup or an exercise for us to get familiarize with the paper, but also an important part of the final results we want to prove. Though not very difficult, having results with both the input and output kernel generalized will give us a much nicer overall package.

The other thing that make this project optimistic is that the estimation after section 4 in the original paper mostly depend on the bound we derive in section 3. More precisely, the limits in lemma 6 and 7 of [1] are the slowest limits we will ever get in this paper. Afterwards all analysis will be done in a higher level. This is huge. If we can reach lemma 6 and lemma 7 then we know that the battle is mostly won. At the end of day, we have a content rich paper where the first half deals with the input kernel and the second half deals with the output kernel. Again, the overall package looks good.

Moreover, this is a data science problem. My other two publications are also in data science, just with more applications. One is about Bayesian learning and variational auto encoder, also a dimension reduction technique like t-SNE. The other one is about learning Markov random fields and k-cnf, also a probabilistic algorithm. Both are published in top tier conferences and we have a theme. Choosing t-SNE as the main project increase the weight of the other two publications because of that. This is a major advantage compared to choosing a statistical physics problem, but I do admit that statistical physics sounds cool.

Finally about the future plans, here are the difficulties I envisioned and what I plan to do.

1. The most difficult part could just well be the first part of the paper, the existence and the uniqueness of σ^* . In [1] the proofs is tailored by the specific form of Gaussian kernels so we might potentially need a brand new proof.
2. The good news is, as long as we can establish the existence of σ^* we don't need the nuances in the proofs for the rest of the problem. We might as well treat it as a separate, small problem to tackle. Moreover, since it is the beginning of the problems it requires minimal prior knowledge, just calculus and dedication. I can ask Tuca to go through the computation with me and ask him for advices.
3. After that there are two routes we can pursue. The first is to compute directly with a generalized kernel. We can write $f(x, x') = e^{w(x, x')}$ because the exponential function is a bijection on $\mathbb{R}^+$. This way we can get cleaner notation and get rid of logarithm as much as

possible. This is the most straightforward and brutal force method. If we can reach lemma 7 in [1] we win.

The other route is the first prove that if a kernel satisfies everything, any kernels that decay faster will also satisfy everything. I asked about this and Tuca firmly believed it is true because it looks way too natural to not be true. If we can do that easily, we then work with a specific kernel and iterate on our choice of specific kernel. Going through this route will potentially be faster, but the result may not be good enough for a thesis.

Either case, here are my tentative timeline.

1. Meet with Tuca again in one week, preferably Monday after he came back from Baltimore. During this week I will explore the existence and the uniqueness of σ^* . I will prepare very detailed notes documenting my effort so when I see Tuca again and if I am having trouble, we can do some computation together.
2. I plan to finish the proof of existence and uniqueness before Christmas, then we can focus on the last part of this project. We will go down both routes and hopefully I will have the problem done in January. Again, we ONLY need to reach lemma 7 so the background is simple enough for Tuca to help and step in. This is very different from the statistical physics problem where it requires too much information Tuca is not familiar with. Plus, come on, Tuca wrote the original paper.
3. After the results are finished, I will put together a carefully reviewed draft for Tuca to review. I will go through it with Tuca and make sure everything is correct.
4. If the results are correct and do not need major repair, I will put together the first draft of the thesis. Tuca may suggest what I have is not enough and needs more. That is OK but a draft is a draft. He and I will discuss and we can also ask opinions from the committee. The goal this time is to make sure the draft I have is well polished so it will not waste any more of my time.
5. If more needs to be done, calm down and discuss with Tuca. If I reach this step, I have solid results with me so the situation is night and day different from before. Tuca will understand and he will help.

3 Setup

Here goes the real math. Recall that this document is dedicated to the reconstruction of [1] with low-dimension kernel g replaced by a more general kernel. This would serve as a great way to understand the material and help with the completion of the original project. For the sake of simplicity, we will use the exact same notation as in [1] and skip all the definitions. Recall in [1], g is defined as

$$g(y, y') = \frac{1}{1 + \|y - y'\|^2} \quad (3.1)$$

With all these being said, we will first start by giving our general low dimension kernel g some general assumptions, and modify it as we move on. We first require that $g(y, y') = k(\|y - y'\|)$ such that:

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing and bounded.
2. k satisfies

$$\iint_{(0, \infty)^2} k(x, x') dx dx' < \infty. \quad (3.2)$$

We will kick off this warmup by listing all the important lemmas and results in order, and prove our own versions when the need arises. The final goal is to prove a same set of results as in [1] with the y kernel replaced. Recall that

$$F_{\rho,\mu}(x, \sigma) = - \int \frac{\|x - x'\|^2}{2\sigma^2} \frac{\exp(-\|x - x'\|^2 / 2\sigma^2)}{\int \exp(-\|x - x'\|^2 / 2\sigma^2) d\mu(x')} d\mu(x') \\ - \log \left(\int \exp(-\|x - x'\|^2 / 2\sigma^2) \right) + \log \rho. \quad (3.3)$$

Proposition 1. *If μ has a C^1 density f such that both f and $\|\nabla f\|$ are bounded, then the equation $F_{\rho,\mu}(x, \sigma) = 0$ has a unique solution $\sigma_{\rho,\mu}^*(x)$ for all $\rho \in (0, 1)$ and $x \in \mathbb{R}^d$. Moreover, $x \mapsto \sigma_{\rho,\mu}^*(x)$ is smooth.*

Proof. This proposition is about the kernel of x . The proof stays unchanged from [1]. $\square$

The next couple of lemmas are general results in section 3 of [1]. We will omit the proof because they are not directly discussing the quantities in the t-sne algorithm. Recall that X or the corresponding probability measure μ is called sub-Gaussian if $\mathbb{P}(X > t) \leq 2 \exp(-Ct^2)$ for some positive constant C .

Lemma 1. *Let ν be sub-Gaussian with second moment equal to 1 and suppose $X \in \mathbb{R}^d$ has i.i.d. entries drawn from ν . Then there exists constant C, c_1 such that*

$$\mathbb{P}(\|X\| \geq \sqrt{d} + t) \leq Ce^{-c_1 t^2}. \quad (3.4)$$

Lemma 2. *Let μ be sub-Gaussian and c_1 be the constant from lemma 1. Let $X_1, \dots, X_n \in \mathbb{R}^d$ have i.i.d. entries drawn from μ . Then there exists a constant C such that*

$$\mathbb{P} \left(\max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\frac{\log n}{c_1}} + \sqrt{d} + t \right) \leq Ce^{-c_1 t^2}. \quad (3.5)$$

Lemma 3. *Let $\delta > 0$ be constant, $K_n = [\delta, \sqrt{n}]$ and $h : \mathbb{R}^d \times \mathbb{R}^d \times \mathbb{R} \rightarrow \mathbb{R}$. Suppose the following hold:*

1. μ is sub-Gaussian;
2. $h(x, x', \sigma)$ and $\nabla_x h(x, x', \sigma)$ are bounded for all x, x' and all $\sigma \in K_n$;
3. $h(x, x', \sigma) \rightarrow 0$ as $\|x - x'\| / \sigma \rightarrow \infty$.

Let $X_1, \dots, X_n$ be i.i.d. random variables drawn from μ . Let $\mu_n = (1/n) \sum_i \delta_{X_i}$. Then for any $\epsilon > 0$ with probability at least $1 - e^{-c\epsilon^2 n}$ we have

$$\sup_{\sigma \in K_n, x} \left| \int h(x, x', \sigma) d\mu_n(x) - \int h(x, x', \sigma) d\mu(x') \right| \leq \epsilon + O \left(\sqrt{\frac{\log n}{n}} \right). \quad (3.6)$$

Define

$$f_\sigma(x, x') = \exp(-\|x - x'\|^2 / 2\sigma^2), \quad g_\sigma = \frac{\|x - x'\|^2}{2\sigma^2} \exp(-\|x - x'\|^2 / 2\sigma^2). \quad (3.7)$$

The next two lemmas use similar techniques to prove.

Lemma 4. *Let c_1 be the decay constant in lemma 1 and let $\delta > 0$. Then almost surely in $X_1, \dots, X_n$,*

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} \left| \frac{\int g_\sigma(X_i, x') d\mu_n(x')}{\int f_\sigma(X_i, x') d\mu_n(x')} - \frac{\int g_\sigma(X_i, x') d\mu(x')}{\int f_\sigma(X_i, x') d\mu(x')} \right| = O(n^{8/c_1 \delta^2 - 1/2} \sqrt{\log n}). \quad (3.8)$$

Lemma 5. Let c_1 be the decay constant in lemma 1 and let $\delta > 0$ be such that $c_1\delta^2 > 8$. Then almost surely in $X_1, \dots, X_n$,

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} \left| \log \left(\int f_\sigma(X_i, x') d\mu_n(x') \right) - \log \left(\int f_\sigma(X_i, x') d\mu(x') \right) \right| = O(n^{4/c_1\delta^2-1/2} \sqrt{\log n}) \quad (3.9)$$

For this pair of lemmas, [1] provides the proof to the first lemma. We will provide the second for the sake of exercise.

Proof of lemma 5. First, notice that for arbitrary $x > 0$, the following elementary inequality holds:

$$\log(x) \leq x - 1. \quad (3.10)$$

Therefore, the lemma can be proved if we are able to show that

$$\begin{aligned} \left| \frac{\int f_\sigma(X_i, x') d\mu_n(x')}{\int f_\sigma(X_i, x') d\mu(x')} - 1 \right| &= \left| \frac{\int f_\sigma(X_i, x') d\mu_n(x') - \int f_\sigma(X_i, x') d\mu(x')}{\int f_\sigma(X_i, x') d\mu(x')} \right| \\ &= O(n^{4/c_1\delta^2-1/2} \sqrt{\log n}). \end{aligned} \quad (3.11)$$

We can achieve that by bounding the numerator and the denominator of the previous formula. On one hand, consider set

$$E'_1 = \left\{ \max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\alpha + \frac{1}{c_1} \log n} + \sqrt{d} \right\} \quad (3.12)$$

Setting $t = \sqrt{\alpha \log n}$ in lemma 2, we have $\mathbb{P}(E'_1) \leq Cn^{-c_1\alpha}$. To absorb the $\sqrt{d}$ term, we can find $\epsilon_n \rightarrow 0$ such that

$$E_1 = \left\{ \max_{i=1, \dots, n} \|X_i\| \geq \sqrt{\alpha + \frac{1}{c_1} + \epsilon_n \log n} \right\} \quad (3.13)$$

with $\mathbb{P}(E_1) \leq Cn^{-c_1\alpha}$. Because the maximum of norm of each X_i is restricted, on E_1^c we have for all i, j , that

$$\|X_i - X_j\|^2 \leq 4 \left(\alpha + \frac{1}{c_1} + \epsilon_n \right) \log n =: 4\gamma \log n. \quad (3.14)$$

Then for $\sigma \in [\delta, \sqrt{n}]$, on E_1^c for all i we have

$$\int f_\sigma(X_i, x') d\mu(x') \geq (1 - Cn^{-c_1\alpha}) \exp \left(-\frac{2\gamma \log n}{\delta^2} \right) \geq \frac{1}{2} \exp \left(-\frac{2\gamma \log n}{\delta^2} \right) = \frac{1}{2} n^{-2\gamma/\delta^2}. \quad (3.15)$$

This helps us bound the denominator. On the other hand, by lemma 3, we are able to find set E_2 with $\mathbb{P}(E_2) \leq e^{-c\epsilon^2 n}$, such that, on E_2^c ,

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} \left| \int f_\sigma(X_i, x') d\mu_n(x') - \int f_\sigma(X_i, x') d\mu(x') \right| \leq \epsilon + O \left(\frac{\sqrt{\log n}}{\sqrt{n}} \right). \quad (3.16)$$

Therefore, we can combine (3.15) and (3.16) and conclude that on $E_1^c \cap E_2^c$,

$$\left| \frac{\int f_\sigma(X_i, x') d\mu_n(x') - \int f_\sigma(X_i, x') d\mu(x')}{\int f_\sigma(X_i, x') d\mu(x')} \right| \leq C\epsilon n^{2\gamma/\delta^2} + O(\sqrt{\log n} \cdot n^{-1/2+2\gamma/\delta^2}). \quad (3.17)$$

In order for the RHS to converge to zero, we need

$$\epsilon n^{2\gamma/\delta^2} \rightarrow 0, \quad 1/2 - 2\gamma/\delta^2 > 0. \quad (3.18)$$

Moreover, we have $\mathbb{P}(E_1 \cup E_2) \leq Cn^{-c_1\alpha} + e^{-c\epsilon^2 n}$. Set $\epsilon = n^{-\eta}$ for some $\eta > 0$. In order to get almost sure convergence using Borel-Cantelli, we also need

$$c_1\alpha > 1, \quad 1 - 2\eta > 0. \quad (3.19)$$

Hence for the entire lemma to work we need the following four inequalities to hold:

$$\frac{2\gamma}{\delta^2} - \eta < 0, \quad \frac{2\gamma}{\delta^2} - 1/2 < 0, \quad c_1\alpha > 1, \quad 1 - 2\eta > 0. \quad (3.20)$$

The last inequality implies that $\eta < 1/2$ so the first inequality implies the second. Now, recall that

$$\gamma := \alpha + \frac{1}{c_1} + \epsilon_n. \quad (3.21)$$

The collection of inequalities is equivalent to

$$2\left(\alpha + \frac{1}{c_1} + \epsilon_n\right) < \eta\delta^2, \quad c_1\alpha > 1, 0 < \eta < 1/2. \quad (3.22)$$

Here, we are free to choose α and η , but c_1 and δ are fixed. Now let

$$\alpha = \frac{1}{c_1} + \epsilon, \quad \eta = \frac{1}{2} - \epsilon. \quad (3.23)$$

Then the second and third inequalities in (3.22) are satisfied. The first is equivalent to

$$\frac{4}{c_1\delta^2} + \epsilon < \frac{1}{2} - \epsilon \quad (3.24)$$

as $\epsilon_n \rightarrow 0$ as $n \rightarrow \infty$. In addition, since ϵ is arbitrary, we have again, equivalently,

$$\frac{4}{c_1\delta^2} - \frac{1}{2} < 0. \quad (3.25)$$

When this is satisfied, we have:

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} \left| \int f_\sigma(X_i, x') d\mu_n(x') - \int f_\sigma(X_i, x') d\mu(x') \right| = O(n^{4/c_1\delta^2-1/2} \sqrt{\log n}). \quad (3.26)$$

The proof is thus finished. □

Then combining the previous result we have an important convergence:

Proposition 2. *Let μ be sub-Gaussian and c_1 be the constant from lemma 1. $X_1, \dots, X_n$ are i.i.d. random variables with probability measure μ and $\mu_n = \frac{1}{n} \sum \delta_{X_i}$ is the empirical measure. Additionally, set $\delta > 0$ to be a constant such that $c_1\delta^2 > 16$. Then, there exists $\epsilon > 0$, such that almost surely in $X_1, \dots, X_n$:*

$$\sup_{\substack{\sigma \in [\delta, \sqrt{n}] \\ i=1, \dots, n}} |F_{\mu_n}(X_i, \sigma) - F_\mu(X_i, \sigma)| = O(n^{-\epsilon}). \quad (3.27)$$

Let us summarize the important results upon this point, which will be used later:

There exists $\gamma = 2/c_1 + \epsilon$ such that the following holds almost surely

- (1) $\max_{i=1,\dots,n} \|X_i\| \leq \sqrt{\gamma \log n}$;
- (2) $\max_{i,j} \|X_i - X_j\|^2 \leq 4\gamma \log n$;
- (3) For $\sigma \in [\delta, \sqrt{n}]$,

$$\int \exp(-\|X_i - x'\|^2 / 2\sigma^2) d\mu_n(x), \int \exp(-\|X_i - x'\|^2 / 2\sigma^2) d\mu(x') \geq Cn^{-2\gamma/\sigma^2}. \quad (3.28)$$

Next couple of lemmas in [1] are about lower bounds on $\sigma_{\rho,\mu}^*$. Recall that $\sigma_{\rho,\mu}^*$ is the unique solution of the equation $F_{\rho,\mu}(x, \sigma_{\rho,\mu}^*) = 0$. For simplicity of notation, we write $\sigma^*(x) = \sigma_{\rho,\mu}^*(x)$ and $\sigma_n^*(x) = \sigma_{\rho,\mu_n}^*(x)$.

Lemma 6. *If μ satisfies the condition in lemma 1 of [1], then $\sigma^*(x)$ and $\sigma_n^*(x)$ are uniformly bounded from below in both i and n almost surely.*

Proposition 3. *Suppose μ has compact support and C^1 density. Then almost surely, as $n \rightarrow \infty$,*

$$\sup_{i=1,\dots,n} |\sigma_n^*(X_i) - \sigma^*(X_i)| \rightarrow 0. \quad (3.29)$$

Proof. By [1], we can quickly reach to the stage where

$$|\sigma_n^*(x) - \sigma^*(x)| = O(|F_{\mu_n}(x, \sigma_n^*(x)) - F_{\mu}(x, \sigma_n^*(x))|) \rightarrow 0. \quad (3.30)$$

Moreover, we know that μ has compact support and $x \mapsto \sigma^*(x)$ is smooth, $\sigma^*(X_i)$ is bounded above and below, meaning that $\sigma_n^*(X_i)$ is bounded uniformly from above and below as well. This completes the proof in the original paper. $\square$

4 More Lemmas

Starting this section we are gradually approaching our point of work. We need to list a few more lemmas before that. To make notations cleaner, [1] introduced some abbreviation in section 4.2. We will use all of them by default.

Lemma 7. *Any sequence in $\tilde{\mathcal{P}}_X$ is tight.*

Recall Prokhorov's theorem:

Theorem (Prokhorov's Theorem). *If μ_n is a tight sequence of measures in Euclidean space. There exists a subsequence μ_{n_k} and a measure μ such that μ_{n_k} converges to μ weakly.*

We immediately get the following lemma:

Lemma 8. *There exists μ^* such that*

$$I_{\rho}(\mu^*) = \inf_{\mu \in \tilde{\mathcal{P}}_X} I_{\rho}(\mu) \quad (4.1)$$

Recall that $I_{\rho}(\mu)$ is the continuous version of $L_{n,\rho}(X, Y)$. It only depends on the measure μ and the perplexity constant ρ . We have not reached the part that requires we have a common perplexity, but from now on we should pay close attention as it is one of the core setup of the original paper.

Lemma 9. As $n \rightarrow \infty$, for some $\epsilon > 0$, we have

$$|f_n(X_i, X_j) - f(X_i, X_j)| = O(n^{-\epsilon}). \quad (4.2)$$

Proof. There are typos in the original proof so we prove this lemmas again despite it being very easy and elementary. We compute directly

$$\begin{aligned} |f_n(x, x') - f(x, x')| &= |\exp(-\|x - x'\|^2 / 2\sigma_n(x)^2) - \exp(-\|x - x'\|^2 / 2\sigma(x)^2)| \\ &= \left| \exp\left(\frac{|\sigma_n(x)^2 - \sigma(x)^2| \|x - x'\|^2}{2\sigma_n(x)^2 \sigma(x)^2}\right) - 1 \right| \cdot \exp(-\|x - x'\|^2 / 2\sigma(x)^2) \\ &\leq \left| \exp\left(\frac{|\sigma_n(x)^2 - \sigma(x)^2| \|x - x'\|^2}{2\sigma_n(x)^2 \sigma(x)^2}\right) - 1 \right|. \end{aligned} \quad (4.3)$$

Recall that the sum of two sub-Gaussian random variables is still sub-Gaussian. If μ_X is sub-Gaussian, let C be the constant of decay for $X_i - X_j$ and we have

$$\mathbb{P}(\|X_i - X_j\|^2 \geq \log n) \leq 2 \exp(-\log n / C^2) = 2 \exp\left(-\frac{1}{C^2}\right) \cdot \frac{1}{n} \rightarrow 0. \quad (4.4)$$

This means $\|X_i - X_j\|^2 = O(\log n)$ almost surely. On the other hand, if μ_X has compact support then $\|X_i - X_j\|$ will be bounded, even easier. Moreover, since σ_n and σ are both uniformly bounded below by lemma (6), by setting $x = X_i$ and $x' = X_j$ we get

$$\begin{aligned} &\left| \exp\left(\frac{|\sigma_n(x)^2 - \sigma(x)^2| \|x - x'\|^2}{2\sigma_n(x)^2 \sigma(x)^2}\right) - 1 \right| \\ &\leq \frac{1}{\sigma_n(x)\sigma(x)} \left(\frac{1}{\sigma_n(x)} + \frac{1}{\sigma(x)} \right) |\sigma_n(x) - \sigma(x)| \|x - x'\|^2 \\ &= O(|\sigma_n(x) - \sigma(x)| \|x - x'\|^2) \\ &= O(|\sigma_n(x) - \sigma(x)|) O(\log n) \\ &= O(n^{-\epsilon}) O(\log n) \\ &= O(n^{-\epsilon}), \end{aligned} \quad (4.5)$$

by proposition (3). This finishes the proof. $\square$

One thing we should highlight here is that we are abusing the notation of σ and σ_n . They are defined as the unique solution to $F = 0$. However, the definition of σ_i is not abused and we will later see why such abuse makes sense.

Lemma 10. If $c_1 \delta > 32$, then there exists $\epsilon > 0$ such that as $n \rightarrow \infty$, we have

$$\begin{aligned} \sup_{i,j} |p'_n(X_i, X_j) - p'(X_i, X_j)| &= O(n^{-\epsilon}), \\ \sup_{i,j} |p_n(X_i, X_j) - p(X_i, X_j)| &= O(n^{-\epsilon}). \end{aligned} \quad (4.6)$$

Supplementary Proof. The overall methodology of the original proof in [1] is natural and clear. However, it can still be slightly confusing as it is missing the actual values we assign to ϵ . In the original paper, we can take, for some constant $k > 0$, that

$$n^{-\epsilon} = n^{8/c_1 \delta^2 - 1/2} (\log n)^k. \quad (4.7)$$

This comes from the detailed computations in the original proof and the quantity above is the slowest decaying speed, the 'worst case scenario' given all the conditions. Plug this into the computations and everything is clear. The proof uses a little bit extra than just the bounds in section 3.1.

□

Lemma 11. *If $c_1\delta > 40$, then there exists $\epsilon > 0$ such that as $n \rightarrow \infty$, we have*

$$\sup_{i,j} |p_n(X_i, X_j) \log p_n(X_i, X_j) - p(X_i, X_j) \log p(X_i, X_j)| = O(n^{-\epsilon}). \quad (4.8)$$

Lemma 12. *If $c_1\delta^2 > 8$, then as $n \rightarrow \infty$ we have*

$$\sum_j p_{j|i} \log p_{j|i} + \log(\rho n) = o(1). \quad (4.9)$$

[Yi: TODO: Ask Tuca about the last step of the proof in [1].]

5 The Actual Computation

From this point onward, we are starting to have lemmas and results related to the output kernel g and this is where the trail begins. Along the way we will see if there is a need to add more conditions to g and if we do, every time we add more constraints we will reiterate the entire settings for g for clarity. To refresh our memory, here is the initial assumptions we have for g : $g(y, y') = k(\|y - y'\|)$ such that:

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing and bounded.
2. k satisfies

$$\iint_{(0,\infty)^2} k(x, x') dx dx' < \infty. \quad (5.1)$$

Lemma 13. *There exists a set of points $y^*(X)$ such that*

$$\inf_{Y \in (\mathbb{R}^s)^n} L_{n,\rho}(X, Y) = L_{n,\rho}(X, y^*). \quad (5.2)$$

[Yi: TODO: Ask Tuca about the $1/n^2$ in the original proof. The scaling seems to not play a role in the proof.]

Proof. Fortunately, the first proof is almost identical to the original one. Let $k \in \{1, \dots, n\}$. We are going to show that

$$\lim_{|y_k| \rightarrow \infty} L_{n,\rho}(X, y) = \infty. \quad (5.3)$$

By definition, L is the relative entropy and

$$L_{n,\rho}(X, y) = \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) = \sum_{i \neq j} p_{ij} \log p_{ij} - \sum_{i \neq j} p_{ij} \log q_{ij}. \quad (5.4)$$

For n given and fixed, it suffices to show that

$$\sum_{i \neq j} p_{ij} \log(q_{ij}) = \sum_{i \neq j} p_{ij} \log \left(\frac{g(y_i, y_j)}{\sum_{m \neq l} g(y_m, y_l)} \right) \rightarrow -\infty. \quad (5.5)$$

Recall that $g(y, y')$ is bounded. We have

$$\begin{aligned} & -\sum_{i \neq j} p_{ij} \log \left(\sum_{m \neq l} g(y_m, y_l) \right) = -\log \left(\sum_{m \neq l} g(y_m, y_l) \right) \\ & \leq -\sum_{i \neq j} p_{ij} \log \left(\sum_{m \neq k, l \neq k, m \neq l} g(y_m, y_l) \right) < \infty. \end{aligned} \quad (5.6)$$

On the other hand, we have $g(y_i, y_k) = k(\|y_i - y_k\|) \rightarrow 0$ as $|y_k| \rightarrow \infty$ because $\|y_i - y_k\| \rightarrow \infty$ and k satisfies

$$\iint_{(0, \infty)^2} k(x, x') dx dx' < \infty. \quad (5.7)$$

This means

$$\lim_{|y_k| \rightarrow \infty} \sum_{i \neq j} p_{ij} \log(g(y_i, y_j)) \rightarrow -\infty. \quad (5.8)$$

Hence, the minimum of the $L_{n, \rho}(X, y)$ is achieved on a bounded set, the lemma follows by the continuity of $L_{n, \rho}(X, y)$. $\square$

Lemma 14. *As $n \rightarrow \infty$, we have*

$$\sum_{i \neq j} p_{ij} \log p_{ij} = \iint p(x, x') \log p(x, x') d\mu d\mu - 2 \log n + o(1). \quad (5.9)$$

Proof. **[Yi: TODO: In the proof we did not use lemma 7 of the original paper. Ask Tuca what did I miss.]** The original paper does provide a detailed proof, so it is our turn to exercise. First recall that $p_{ij} = p_n(X_i, X_j)/n^2$. Therefore we have

$$\begin{aligned} \sum_{i \neq j} p_{ij} \log ij &= \sum \frac{1}{n^2} p_n(X_i, X_j) \log \left(\frac{1}{n^2} p_n(X_i, X_j) \right) \\ &= \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log(p_n(X_i, X_j)) - 2 \log n \sum_{i \neq j} \frac{1}{n^2} p_n(X_i, X_j) \\ &= \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log(p_n(X_i, X_j)) - 2 \log n, \end{aligned} \quad (5.10)$$

because the second summation is just the sum of all probability mass. Now we have the $2 \log n$ term and we are left to deal with the first summation. Inspired by the proofs in [1], we begin by showing that the diagonal terms do not matter in our estimation. consider

$$\begin{aligned} & \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log(p_n(X_i, X_j)) - \iint p_n(x, x') \log p_n(x, x') d\mu_n(x) d\mu_n(x') \\ &= -\frac{1}{n^2} \sum_{t=1}^n p'_n(X_t, X_t) \\ &= -\frac{1}{n^2} \sum_{t=1}^n \frac{1}{\int f(X_t, x') d\mu_n(x') - \frac{1}{n}} \log \left(\frac{1}{\int f(X_t, x') d\mu_n(x') - \frac{1}{n}} \right). \end{aligned} \quad (5.11)$$

If we refer to the proof of lemma 14 in [1], we can see that the difference of diagonal term vanishes and it does not depend on the specific X_i . Therefore in our proof, when we sum

over all the X_i and normalize by dividing n^2 , we reach the exact same conclusion. The diagonal term is $O(n^{4/c_1\delta^2-1} \log n)$ which is $o(1)$ depending on our choice of c_1 .

Then we apply lemma 11 and since $d\mu_n \times d\mu_n$ is a probability measure, we have

$$\iint p_n(x, x') \log p_n(x, x') d\mu_n(x) d\mu_n(x') = \iint p(x, x') \log p(x, x') d\mu_n(x) d\mu_n(x') + o(1), \quad (5.12)$$

which immediately implies

$$\sum_{i \neq j} p_{ij} \log ij = \iint p(x, x') \log p(x, x') d\mu_n(x) d\mu_n(x') - 2 \log n + o(1). \quad (5.13)$$

The proof will be finished if we are able to replace the measure in the last integral with $d\mu$. We verify that

$$h(x, x', \sigma) := p(x, x') \log p(x, x') \quad (5.14)$$

satisfies all the requirements in lemma 3. First μ is sub-Gaussian by construction. Then is easy to verify that h is C^1 for all x, x' and σ . Moreover, by calculus we have $p(x, x') \in (0, 1]$ and $\lim_{a \rightarrow 0} a \log a = 0$. Thus $p \rightarrow 0$ as $\|x - x'\|/\sigma \rightarrow \infty$. It is also obvious that h is bounded and $h \rightarrow 0$ as $\|x - x'\|/\sigma \rightarrow \infty$. The rest of the conditions are also very easily to prove. We can finally conclude that

$$\iint p(x, x') \log p(x, x') d\mu_n(x) d\mu_n(x') = \iint p(x, x') \log p(x, x') d\mu(x) d\mu(x') + o(1). \quad (5.15)$$

Combine this estimation with (5.13) and we can conclude

$$\sum_{i \neq j} p_{ij} \log ij = \iint p(x, x') \log p(x, x') d\mu(x) d\mu(x') - 2 \log n + o(1), \quad (5.16)$$

as desired. □

Partial Derivative of the Relative Entropy

Before we move on, we would like to take a small deviation and compute the partial derivative of the relative entropy. This is important because it will impact the property of the set $\tilde{\mathcal{P}}_X$.

Lemma 15. *Let $X \in (\mathbb{R}^d)^n$ and $Y = (y_1, \dots, y_n) \in (\mathbb{R}^s)^n$ be the input and output of the t -SNE algorithm, then we have*

$$\frac{\partial L_{n,\rho}(X, Y)}{\partial y_i} = - \sum_{j=1, j \neq i}^n (p_{ij} - q_{ij}) \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)}. \quad (5.17)$$

Proof. By definition,

$$\begin{aligned} L_{n,\rho}(X, Y) &= \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) \\ &= \sum_{i \neq j} p_{ij} \log p_{ij} - \sum_{i \neq j} p_{ij} \log q_{ij}. \end{aligned} \quad (5.18)$$

The first summation does not depend on y_i and will vanish when taking the partial derivative. Moreover, we can further breakdown the second summation:

$$\begin{aligned} - \sum_{i \neq j} p_{ij} \log q_{ij} &= - \sum_{i \neq j} p_{ij} \log g(y_i, y_j) + \sum_{i \neq j} p_{ij} \log \left(\sum_{k \neq l} g(y_k, y_l) \right) \\ &= - \sum_{i \neq j} p_{ij} \log g(y_i, y_j) + \log \left(\sum_{k \neq l} g(y_k, y_l) \right). \end{aligned} \quad (5.19)$$

Taking derivative of the last expression and note that only terms that contain y_i remains. This means

$$\begin{aligned} \frac{\partial L_{n,\rho}(X, Y)}{\partial y_i} &= \sum_{j \neq i, j=1}^n \left(-p_{ij} \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)} + \frac{1}{\sum_{k \neq l} g(y_k, y_l)} \frac{\partial}{\partial y_i} g(y_i, y_j) \right) \\ &= \sum_{j \neq i, j=1}^n \left(-p_{ij} \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)} + \frac{g(y_i, y_j)}{\sum_{k \neq l} g(y_k, y_l)} \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)} \right) \\ &= - \sum_{j=1, j \neq i}^n (p_{ij} - q_{ij}) \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)}. \end{aligned} \quad (5.20)$$

□

We can easily do a sanity test by letting $g(y_i, y_j) = (1 + \|y_i - y_j\|^2)^{-1}$ and get formula (A.1) in [1].

Back to the Main Arena

The derivative in lemma 15 explains why $\tilde{\mathcal{P}}_X$ has that specific form. We are gradually marching towards the end. Onto the next lemma we go:

Lemma 16. *The following holds:*

$$\liminf_{n \rightarrow \infty} \sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) \geq \iint p(x, x') \log \frac{p(x, x')}{q(y, y')} d\mu d\mu. \quad (5.21)$$

Proof. Good news, the proof of this lemma also requires minimal details of the output kernel and is just a simple application of the Fatou's lemma. Like in lemma 15, we first break down the summation:

$$\sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) = \sum_{i \neq j} p_{ij} \log p_{ij} - \sum_{i \neq j} p_{ij} \log q_{ij}. \quad (5.22)$$

The convergence of the first summation has been proved in lemma 14. We just need to deal with the second summation. Note that we now have

$$\mu_n = \frac{1}{n} \sum \delta_{X_i} \times \delta_{y_i} \quad (5.23)$$

if y_i is involved. By the definition of q_{ij} , we have

$$\begin{aligned}
& - \sum_{i \neq j} p_{ij} \log(q_{ij}) \\
&= \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log g(y_i, y_j) + \log \left(\frac{1}{n^2} \sum_{i \neq j} g(y_i, y_j) \right) + 2 \log n \\
&= \iint p_n(x, x') g(y, y') d\mu_n d\mu_n + \log \left(\iint g(y, y') d\mu_n d\mu_n - \frac{1}{n} \right) + 2 \log n. \tag{5.24}
\end{aligned}$$

By Fatou's lemma and lemma 10, we have

$$\liminf_{n \rightarrow \infty} \iint p_n(x, x') \log g(y, y') d\mu_n d\mu_n \geq \iint p(x, x') \log g(y, y') d\mu d\mu. \tag{5.25}$$

Similarly, using Fatou lemma again and we have

$$\liminf_{n \rightarrow \infty} \iint g(y, y') d\mu_n d\mu_n - \frac{1}{n} \geq \iint g(y, y') d\mu d\mu. \tag{5.26}$$

The $1/n$ is the duplicated diagonal term in the empirical measure and will obviously vanish as we take $n \rightarrow \infty$. Moreover, because of our assumption on the kernel g , the integrals on the RHS of the last two inequalities converge. In addition, since $\log$ is a continuous and increasing function, (5.26) implies

$$\liminf_{n \rightarrow \infty} \log \left(\iint g(y, y') d\mu_n d\mu_n - \frac{1}{n} \right) \geq \log \left(\iint g(y, y') d\mu d\mu \right). \tag{5.27}$$

The proof is finished by combining (5.24), (5.25), (5.27) and lemma 14. The $2 \log n$ cancelled out because of (5.13). □

In the original paper, this is the point where we finish proving the lower bound of the main theorem. However, we need to postpone such triumph later as we need a later lemma to reconstruct the results in the appendix of [1]. For convenience, define

$$q_n(y, y') = \frac{g(y, y')}{\iint g(y, y') d\mu_n(y) d\mu_n(y') - \frac{1}{n} k(0)}. \tag{5.28}$$

Here $k(0) = g(y, y)$ is the value of the output kernel when both inputs are equal. Since multiplying a constant to g does not change anything, we can just treat $k(0) = 1$ for convenience as everything is normalized. By lemma 8 there exists μ^* such that $I(\mu^*) = \inf_{\mu} I_{\rho}(\mu)$. Let $(X_1, Y_1), \dots, (X_n, Y_n)$ be i.i.d. random variables drawn from μ^* and $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{(X_i, Y_i)}$ their empirical measure. μ_n converges to μ^* weakly.

Lemma 17. *As $n \rightarrow \infty$, we have*

$$\sum_{i \neq j} \log \left(\frac{p_{ij}}{q_{ij}} \right) = \iint p_n(x, x') \log \left(\frac{p_n(x, x')}{q_n(y, y')} \right) d\mu_n d\mu_n + o(1). \tag{5.29}$$

Proof. We begin with the familiar computation

$$\begin{aligned}
\sum_{i \neq j} p_{ij} \log \left(\frac{p_{ij}}{q_{ij}} \right) &= \frac{1}{n^2} \sum_{i \neq j} p_n(X_i, X_j) \log \left(\frac{p_n(X_i, X_j)}{q_n(X_i, X_j)} \right) \\
&= \iint p_n(x, x') \log \left(\frac{p_n(X_i, X_j)}{q_n(X_i, X_j)} \right) d\mu_n d\mu_n + \text{Err}_n, \tag{5.30}
\end{aligned}$$

where

$$\begin{aligned}
& \text{Err}_n \\
&= -\frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \left(\log \left(\frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \right) - \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq l} g(Y_k, Y_l)} \right) \right) \\
&= -\frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \right) \\
&\quad + \frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq l} g(Y_k, Y_l)} \right). \tag{5.31}
\end{aligned}$$

[Yi: Ask Tuca about the errors in the original paper.] By (4.4), we know that $\|X_i - X_j\| \leq \log n$ almost surely as $n \rightarrow \infty$. This means $f_n(X_i, X_k) \geq 1/n + \exp(-2\sigma_n)$ almost surely. Using this fact it is easy to see that the first summation in the error term is $o(1)$ and we can write

$$\text{Err}_n = o(1) + \frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} \log \left(\frac{1}{\frac{1}{n} \sum_{k \neq l} g(Y_k, Y_l)} \right). \tag{5.32}$$

For the second term, it is already clear that

$$\frac{1}{n^2} \sum_{i=1}^n \frac{1}{\frac{1}{n} \sum_{k \neq i} f_n(X_i, X_k)} = o(1). \tag{5.33}$$

On the other hand, we have by Fatou's lemma:

$$\liminf_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k \neq l} g(Y_k, Y_l) = \liminf_{n \rightarrow \infty} \int \int g(y, y') d\mu_n d\mu_n - \frac{1}{n} \geq \int \int g(y, y') d\mu^* d\mu^* < \infty, \tag{5.34}$$

by our assumption on the output kernel g . We can see that the second term in Err_n is also $o(1)$, which immediately implies $\text{Err}_n = o(1)$ and completes the proof. $\square$

The next lemma is needed to closing up the lower bound part of the main theorem.

Lemma 18. *As $n \rightarrow \infty$, we have*

$$\int \int g(y, y') d\mu_n(y) d\mu_n(y) - \frac{1}{n} = \int \int g(y, y') d\mu^* d\mu^* + o(1). \tag{5.35}$$

Proof. In this proof we will be adding some minor constraints to $g(y, y')$ to make the proof work. Since $\mu_n \rightarrow \mu$ weakly, it is tight by Prokhorov's theorem. For arbitrary $\epsilon > 0$, there exists a compact set K such that for all n , $\mu^*(K^c), \mu_n(K^c) \leq \epsilon$. **[Yi: TODO: Ask Tuca about the typo in the original paper.]** Since $g(y, y')$ is bounded, there exists a constant C such that

$$\begin{aligned}
& \left| \int \int g(y, y') d\mu_n(y) d\mu_n(y) - \int \int g(y, y') d\mu^* d\mu^* \right| \\
&= \left| \int \int g(y, y') \mathbb{1}_{K \times K} (d\mu_n d\mu_n - d\mu^* d\mu^*) \right| + C\epsilon. \tag{5.36}
\end{aligned}$$

Define

$$G_n(y) = \int g(y, y') \mathbb{1}_K d\mu_n(y'), \quad G(y) = \int g(y, y') \mathbb{1}_K d\mu^*(y'). \tag{5.37}$$

We now require the kernel g to have bounded derivative, and we have the following claims:

Claim 1. $G_n(y)$ is uniformly equicontinuous on K .

Proof of Claim. Recall that by our assumptions $g(y, y') = k(\|y - y'\|)$. We get

$$\begin{aligned} |G_n(y) - G_n(\tilde{y})| &= \left| \int k(\|y - y'\|) - k(\|\tilde{y} - y'\|) \mathbb{1}_K d\mu_n(y') \right| \\ &\leq \int \|y - \tilde{y}\| \left| \frac{k(\|y - y'\|) - k(\|\tilde{y} - y'\|)}{\|(y - y') - (\tilde{y} - y')\|} \right| \mathbb{1}_K d\mu_n(y') \\ &= O(\|y - \tilde{y}\|). \end{aligned} \quad (5.38)$$

by mean value theorem and the additional assumption we just added. $\square$

Claim 2. We have $\sup_{y \in K} |G_n(y) - G(y)| = o(1)$.

Proof of Claim. Since K is compact, it suffices to show that for each $y \in K$, there exists $\delta > 0$ such that

$$\sup_{\tilde{y} \in B(y, \delta)} |G_n(\tilde{y}) - G(\tilde{y})| \leq \epsilon. \quad (5.39)$$

[Yi: TODO: Ask Tuca how exactly compactness plays a role in the proof. Using the finite subcover of all the balls?] Towards this end, by triangular inequality,

$$|G_n(\tilde{y}) - G(\tilde{y})| \leq |G_n(\tilde{y}) - G_n(y)| + |G_n(y) - G(y)| + |G(y) - G(\tilde{y})|. \quad (5.40)$$

By the previous claim, the first term on the RHS vanishes because of equicontinuity. The second term vanishes because of weak convergence and the boundedness of the integrand $g(y, y')$. Finally, the last term vanishes because we can prove the equicontinuity of G using a similar argument like the claim 1. $\square$

By claim 2 we have

$$\iint g(y, y') \mathbb{1}_{K \times K} d\mu_n d\mu_n = \iint g(y, y') \mathbb{1}_{K \times K} d\mu^* d\mu_n + o(1) = \int G(y) \mathbb{1}_K d\mu_n(y) + o(1). \quad (5.41)$$

Since G is bounded and continuous by our assumptions on the kernel $g(y, y')$, using weak convergence of μ_n and we have

$$\int G(y) \mathbb{1}_K d\mu_n(y) = \int G(y) \mathbb{1}_K d\mu^*(y) + o(1) = \iint g(y, y') d\mu^* d\mu^* + o(1), \quad (5.42)$$

as desired. $\square$

For clarify, we reiterate our assumptions: The output kernel satisfies $g(y, y') = k(\|y - y'\|)$ such that:

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing and bounded.
2. k satisfies

$$\iint_{(0, \infty)^2} k(x, x') dx dx' < \infty. \quad (5.43)$$

3. k has bounded derivative.

We are near the end, a couple of more lemmas to go!

Lemma 19. *If μ^* has compact support in Y , then*

$$\iint p_n(x, x') \log(g(y, y')) d\mu_n d\mu_n = \iint p_n(x, x') \log(g(y, y')) d\mu^* d\mu^* + o(1). \quad (5.44)$$

[Yi: TODO: Ask Tuca what does he mean by 'having compact support in Y '. Y is the output data we are trying to compute, not a space.]

Proof. It seems that the compactness is used to ensure the upper bound of $\|y - y'\|$ such that $g(y, y')$ does not decay to zero. In that case, by lemma 10, we can replace the p_n in the integral by p :

$$\iint p_n(x, x') \log(g(y, y')) d\mu_n d\mu_n = \iint p(x, x') \log(g(y, y')) d\mu_n d\mu_n + o(1). \quad (5.45)$$

mimicking the proof of lemma 14 we can show that

$$\iint p(x, x') \log(g(y, y')) d\mu_n d\mu_n = \iint p(x, x') \log(g(y, y')) d\mu^* d\mu^* + o(1). \quad (5.46)$$

This finishes the proof. □

Lemma 20. *If μ^* has compact support in Y , then*

$$\sum_{i \neq j} p_{ij} \log\left(\frac{p_{ij}}{q_{ij}}\right) = \iint p(x, x') \log\left(\frac{p(x, x')}{q(y, y')}\right) d\mu^* d\mu^* + o(1) \quad (5.47)$$

Proof. This lemma can be proved by combining the last 3 lemmas and lemma 14. □

Using the previous lemma and we can immediately get **[Yi: TODO: I maybe stupid but I need to ask Tuca about this.]**

Proposition 4. *If μ^* has compact support in Y , then*

$$\limsup_{n \rightarrow \infty} \inf_{Y \in (\mathbb{R}^s)^n} L_{n, \rho}(X, Y) \leq I_\rho(\mu^*). \quad (5.48)$$

6 Finishing up the Lower Bound

Now with lemma 18 proved, we are able to tackle the lemmas in the appendix and prove our own versions of them. Recall that $\mu_n = \frac{1}{n} \sum_{i=1}^n \delta_{X_i, Y_i}$ is the empirical measure of both the input and output. We just ignore the first or the second part based on the integration variable because it is a probability measure. Let $Y = Y^* = (Y_1, \dots, Y_n)$ be a minimizer of L . Recall from (5.20) that

$$\frac{\partial L_{n, \rho}(X, Y)}{\partial y_i} = - \sum_{j=1, j \neq i}^n (p_{ij} - q_{ij}) \frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)} = 0. \quad (6.1)$$

Lemma 21. *Suppose that μ_X has either compact support and a C^1 density $f(x)$ or μ_X has sub-Gaussian tails. Then for μ -a.e. x and y we have*

$$\int (p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu(x', y') = 0. \quad (6.2)$$

Proof. Let $\alpha(x, y)$ be a test function. It suffices to show that

$$\iint \alpha(x, y)(p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu(x', y') = 0. \quad (6.3)$$

Recalling the definition of p_n and q_n , we see from (5.20) that for all i ,

$$\frac{1}{n^2} \sum_{j=1, j \neq i}^n (p_n(X_i, X_j) - q_n(X_i, X_j)) \frac{\frac{\partial}{\partial Y_i} g(Y_i, Y_j)}{g(Y_i, Y_j)} = 0. \quad (6.4)$$

Multiply by $\alpha(X_i, Y_i)$ for each i and the both sides are still zero. Afterwards, take summation in i and we get

$$\begin{aligned} & \iint \alpha(x, y)(p_n(x, x') - q_n(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu_n d\mu_n \\ &= \frac{1}{n^2} \sum_{1 \leq i, j \leq n} (p_n(X_i, X_j) - q_n(X_i, X_j)) \frac{\frac{\partial}{\partial Y_i} g(Y_i, Y_j)}{g(Y_i, Y_j)} = 0 \end{aligned} \quad (6.5)$$

if we require $k'(0) = 0$. By lemma 10 and lemma 18, we have

$$p_n(X_i, X_j) - q_n(Y_i, Y_j) = p(X_i, X_j) - q(Y_i, Y_j) + o(1). \quad (6.6)$$

Since both α and $\frac{\frac{\partial}{\partial y_i} g(y_i, y_j)}{g(y_i, y_j)}$ are bounded based on our assumptions, we have

$$\begin{aligned} & \iint \alpha(x, y)(p_n(x, x') - q_n(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu_n d\mu_n \\ &= \iint \alpha(x, y)(p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu_n d\mu_n + o(1). \end{aligned} \quad (6.7)$$

By lemma 22 and lemma 23 below we have

$$\begin{aligned} & \iint \alpha(x, y)(p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu_n d\mu_n \\ &= \iint \alpha(x, y)(p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu d\mu + o(1). \end{aligned} \quad (6.8)$$

Since α is an arbitrary function,

$$\int (p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu(x', y') = 0. \quad (6.9)$$

□

Let us now update the assumptions we have on $g(y, y')$: The output kernel satisfies $g(y, y') = k(\|y - y'\|)$ such that:

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing and bounded.
2. k satisfies

$$\iint_{(0, \infty)^2} k(x, x') dx dx' < \infty. \quad (6.10)$$

3. k has bounded derivative.
4. $k'(0) = 0$.

Lemma 22. *The following holds as $n \rightarrow \infty$:*

$$\iint \alpha(x, y) p(x, x') \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu_n d\mu_n = \iint \alpha(x, y) p(x, x') \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu d\mu + o(1) \quad (6.11)$$

This lemma can be proved by mimicking lemma 14

Lemma 23. *The following holds as $n \rightarrow \infty$:*

$$\iint \alpha(x, y) q(y, y') \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu_n d\mu_n = \iint \alpha(x, y) q(y, y') \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu d\mu + o(1) \quad (6.12)$$

This lemma can be proved by mimicking lemma 17. Now let us go back to lemma 13. Let y^* be given in the lemma and define

$$\mu_n = \frac{1}{n} \sum \delta_{X_i} \times \delta_{y_i}. \quad (6.13)$$

We can easily show that μ_n converges weakly to a measure μ . By lemma 21, we know that the limiting measure $\mu \in \tilde{\mathcal{P}}_X$. Then, using lemma 13 and lemma 16, we have the following important proposition:

Proposition 5. *There exists a sub-sequence $n(k)$ such that*

$$\liminf_{k \rightarrow \infty} \inf_{Y \in (\mathbb{R}^s)^{n(k)}} L_{n(k), \rho}(X, Y) \geq I_\rho(\mu). \quad (6.14)$$

7 Proof of the Main Theorems

We now state the main theorems and prove it.

Theorem 1. *[Yi: TODO: Ask Tuca about the sub-Gaussian tails.] Assume μ_X has C^1 density $f(x)$ and compact support. Let $X_i \in \mathbb{R}^d$ be independent, identically distributed random variables with common distribution μ_X . Then for any $\rho \in (0, 1)$,*

$$\lim_{n \rightarrow \infty} \inf_Y L_{n, \rho}(X, Y) = \inf_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu). \quad (7.1)$$

Moreover, there exists a measure μ^* and a sub-sequence n_k such that μ_{n_k} converges weakly to μ^* and $I_\rho(\mu^*) = \inf_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu)$

Proof. Write

$$d_{n, \rho}(X) = \inf_{Y \in (\mathbb{R}^s)^n} L_{n, \rho}(X, Y). \quad (7.2)$$

By proposition 4, 5 and the fact that $\mu^* = \arg \min_{\mu \in \tilde{\mathcal{P}}_X} I_\rho(\mu)$, there exists $\mu \in \tilde{\mathcal{P}}_X$ and a subsequence $n(k)$ such that

$$I_\rho(\mu^*) \geq \limsup_{n \rightarrow \infty} d_{n, \rho}(X) \geq \limsup_{n \rightarrow \infty} d_{n(k), \rho}(X) \geq I_\rho(\mu) \geq I_\rho(\mu^*). \quad (7.3)$$

Therefore the limit

$$\lim_{n \rightarrow \infty} d_{n(k), \rho}(X) = I_\rho(\mu) = I_\rho(\mu^*) \quad (7.4)$$

holds when we look at the sub-sequence $n(k)$ with the empirical measure μ converging to μ^* weakly. For the general limit, we can repeat the reasoning above by looking at any convergent sub-sequences of $d_{n, \rho}(X)$. We are able to find a sub-sequence inside each convergent sub-sequence and they all converges to $I_\rho(\mu^*)$. Therefore each convergent sub-sequence of $d_{n, \rho}(X)$ has the same limit and the theorem is proved. $\square$

Theorem 2. Under the same assumptions of theorem 1, μ^* has compact support.

We need another lemma before proving this theorem.

Lemma 24. For every $\mu \in \tilde{\mathcal{P}}_X$, there exists a function $h_\mu(x)$ that does depend on y such that *[Yi: I think h_μ does NOT depend on y as its partial derivative with respect to y is zero by the deduction below? Ask Tuca if it is a typo.]*

$$\int p(x, x') \log(1 + \|y - y'\|) d\mu = h_{\mu(x)} - \frac{\int g(y, y') d\mu}{\iint g(y, y') d\mu d\mu}. \quad (7.5)$$

Moreover, if μ_X has compact support, then h_μ is continuous on μ_X .

Proof. Recall from (15), we have

$$\begin{aligned} 0 &= \int (p(x, x') - q(y, y')) \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu(x', y') \\ &= \int p(x, x') \frac{\frac{\partial}{\partial y} g(y, y')}{g(y, y')} d\mu - \frac{\int \frac{\partial}{\partial y} g(y, y') d\mu}{\iint g(y, y') d\mu d\mu} \\ &= \nabla_y \left(\int p(x, x') \log(g(y, y')^{-1}) d\mu - \frac{\int g(y, y') d\mu}{\iint g(y, y') d\mu d\mu} \right). \end{aligned} \quad (7.6)$$

This means that there exists $h_\mu : \mathbb{R}^d \rightarrow \mathbb{R}$ such that

$$h_{\mu(x)} = \int p(x, x') \log(g(y, y')^{-1}) d\mu + \frac{\int g(y, y') d\mu}{\iint g(y, y') d\mu d\mu}. \quad (7.7)$$

Since $h_{\mu(x)}$ does not change with y , it suffices to show that $\int p(x, x') \log(g(y, y')^{-1}) d\mu$ is continuous in x . Let $\epsilon > 0$ and suppose $x_n \rightarrow x$. Since p is continuous, and μ_X has compact support, we have for some constant C ,

$$\iint \log(g(y, y')^{-1}) d\mu d\mu \leq C \int p(x, x') \log(g(y, y')^{-1}) d\mu < \infty. \quad (7.8)$$

In particular, for almost every y , $\int \log(g(y, y')^{-1}) d\mu < \infty$. Since $p(x, x')$ is continuous and μ_X has compact support, $p(x, x')$ is uniformly continuous. Specifically, there exists $\delta = \delta(y, \mu)$ such that if $\|x_n - x\| \leq \delta$,

$$\sup_x |p_n(x, x') - p(x, x')| \leq \frac{\epsilon}{\log(g(y, y')^{-1}) d\mu}. \quad (7.9)$$

This means

$$\left| \int (p_n(x, x') - p(x, x')) \log(g(y, y')^{-1}) d\mu \right| < \epsilon, \quad (7.10)$$

which is the definition of continuity. The proof is finished. $\square$

Proof of Theorem 2. We prove by contradiction. Define $D_n = \{\|y\| \geq n\}$. If μ^* does not have compact support in Y then $\mu(D_n) > 0$ for all n . By lemma 24, there exists continuous $h(x)$ such that

$$\int p(x, x') \log(g(y, y')^{-1}) d\mu^* = h(x) - \frac{\int g(y, y') d\mu^*}{\iint g(y, y') d\mu^* d\mu^*}. \quad (7.11)$$

Multiply both sides by $\mathbb{1}_{D_n(x,y)}/\mu^*(D_n)$ and integrate with respect to the other variable:

$$\begin{aligned} & \iint p(x, x') \log(g(y, y')^{-1}) \frac{\mathbb{1}_{D_n(x,y)}}{\mu^*(D_n)} d\mu^* d\mu^* \\ &= \int h(x) \frac{\mathbb{1}_{D_n(x,y)}}{\mu^*(D_n)} d\mu^* - \frac{\iint g(y, y') \frac{\mathbb{1}_{D_n(x,y)}}{\mu^*(D_n)} d\mu^* d\mu^*}{\iint g(y, y') d\mu^* d\mu^*}. \end{aligned} \quad (7.12)$$

Since h is continuous and μ_X has compact support, the first term of the RHS is bounded uniformly in n . Moreover, $g(y, y')$ is bounded so the second term is also bounded uniformly in n . It follows that the LHS is bounded in uniformly in n . With this being said, because $p(x, x') > 0$ and μ_X has compact support, there exists a constant c such that $p(x, x') \geq c$ for all x, x' in the support of μ_X .

Meanwhile, since the output kernel $g(y, y') = k(\|x - x'\|)$ is integrable, it has a decaying tail and there exists $C(n)$ such that $C(n) \rightarrow \infty$ and if $\|x - x'\| \geq n/2$, $g(y, y')^{-1} = k(\|x - x'\|)^{-1} \geq \log(1 + C(n))$. We compute

$$\begin{aligned} & \frac{1}{\mu^*(D_n)} \iint p(x, x') \log(g(y, y')^{-1}) \mathbb{1}_{\|y\| \geq n} d\mu^* d\mu^* \\ & \geq \frac{1}{\mu^*(D_n)} \iint p(x, x') \log(g(y, y')^{-1}) \mathbb{1}_{\|y\| \geq n} \mathbb{1}_{\|y - y'\| \geq n/2} d\mu^* d\mu^* \\ & \geq c \frac{\log(1 + C(n))}{\mu^*(D_n)} \iint \mathbb{1}_{\|y\| \geq n} \mathbb{1}_{\|y'\| \leq n/2} d\mu^* d\mu^* \\ & = c \log(1 + C(n)) \mu^*(\|Y'\| \leq n/2). \end{aligned} \quad (7.13)$$

Clearly,

$$\lim_{n \rightarrow \infty} c \log(1 + C(n)) \mu^*(\|Y'\| \leq n/2) = \infty. \quad (7.14)$$

This contradicts with the boundedness. We must have that $\mu^*(D_n) = 0$ for n sufficiently large. That is, $\mu^*(\|Y\| \geq n) = 0$, so μ^* has compact support in Y . $\square$

In the end, we have the following assumptions on the output kernel for everything to hold: The output kernel satisfies $g(y, y') = k(\|y - y'\|)$ such that:

1. $k : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is decreasing and bounded.
2. k satisfies

$$\iint_{(0, \infty)^2} k(x, x') dx dx' < \infty. \quad (7.15)$$

3. k has bounded derivative.
4. $k'(0) = 0$.

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